# Muhammad Mahasin Irfan

AI Security and Privacy

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## Research Summary

I am an undergraduate researcher at LUMS interested in security and privacy, especially how we measure AI systems and how people handle security-critical technologies in practice. My human-centered work is supervised by researchers at LUMS, CISPA, and ICSI. I also build experimental systems and evaluation methods for machine learning research. I work at the Security and Internet Analytics Lab and the Internet Security and Privacy Lab at LUMS.

## Research Interests

Security and privacy; trustworthy measurement; machine learning systems; empirical evaluation.

## Education

**Lahore University of Management Sciences (LUMS)** Lahore, Pakistan
*B.S. in Computer Science; CGPA: 3.85/4.00* *September 2023 – August 2027 (expected)*

- Dean's Honor List, 2023–24, 2024–25, and 2025–26.
- Selected coursework: Operating Systems; Computer Systems; Networks and Systems; Data Structures; Foundations of Artificial Intelligence and Machine Learning.

## Publications

**Accepted**

- **M. M. Irfan**, M. S. Hassan, and F. Zaffar. "Decomposing Error in Continual World Models." NeurIPS 2026 Workshop on Continual World Models, Idea Track. **Accepted** First author.

**In preparation**

- **M. M. Irfan**$^*$ et al. Security and Privacy Intermediaries in Mobile-Phone Shops (working title). Joint first author ($^*$equal contribution).
- Heterogeneous systems manuscript on optimal ML workload splitting across heterogeneous memory controllers (GPU and processing-in-memory).

## Research Experience

**Decomposing Error in Continual World Models** July 2026 – Present
*First-Author Researcher, LUMS; NeurIPS 2026 Workshop on Continual World Models (Idea Track)* *Lahore, Pakistan*
*Supervised by Dr. Fareed Zaffar (LUMS)*

- Derived an exact decomposition of post-update prediction error into inherited error and update mismatch, showing why endpoint accuracy alone may mis-rank continual model updates.
- Demonstrated opposite endpoint/mismatch rankings in 544/544 controlled one-step trials; the reversal persisted in 93.8% of trials at horizon 5 and 66.2% at horizon 12.
- Reproduced the phenomenon across 500 learned-model runs on a nonlinear damped-pendulum simulator; reversal frequency fell from 95% to 6% as inherited error decreased from 0.038 to 0.004.
- Developed the mathematical formulation, implemented the experiments, analyzed the results, and wrote the first-author manuscript.

**Security and Privacy Intermediaries in Mobile-Phone Shops** June 2026 – Present
*Joint-First-Author Researcher, LUMS/CIRCLE; collaboration with CISPA* *Lahore, Pakistan*
*Supervised by Dr. Mobin Javed (LUMS), Dr. Katharina Krombholz (CISPA), and Dr. Lea Grober (formerly CISPA, ICSI Berkeley)*

- Co-designed an IRB-approved qualitative study of mobile-phone shopkeepers as security and privacy intermediaries for low-income communities in Pakistan.
- Conducted 15 semi-structured Urdu interviews in Lahore on account creation and recovery, device migration, authentication, privacy advice, and fraud; each interview lasted approximately 30 minutes.
- Transcribed, translated, and coded interviews and conducted collaborative reflexive thematic analysis.

**Optimal ML Workload Splitting Across Heterogeneous Memory Controllers** November 2025 – Present

*Undergraduate Researcher, LUMS; collaboration with CMU, UIUC, and AMD Research* *Lahore, Pakistan*

*Supervised by Dr. Fareed Zaffar (LUMS), Dr. Hashim Sharif (PhD, UIUC; AMD Research), and Humza Ikram (CMU PhD student)*

- Created a trustworthy measurement pipeline for latency and power to support the experiments.
- Developing efficient layer-partitioning methods for heterogeneous GPU and processing-in-memory execution, evaluated on four CNNs and four LLMs ranging from MobileNetV2 to Qwen3-32B.
- Extended a Newton PIM model in Ramulator using C++ to explore additional bank groups and larger row sizes with corresponding $t_{FAW}$ adjustments, improving simulated PIM throughput by 1.3× and 1.5×, respectively, over the original configuration.
- Proposed a prediction-guided split-search heuristic that evaluates up to 7–8× fewer candidates while selecting configurations whose measured end-to-end latency is within 1% of an exhaustive oracle in the best evaluated cases.

**Quantizing Diffusion Models** June 2025 – August 2025

*AI Optimization Intern, 10x Engineers* *Lahore, Pakistan*

*Supervised by Dr. Saad Bin Nasir (10xEngineers; formerly Qualcomm) and Dr. Khurram Usman Mazher (formerly Qualcomm)*

- Extended QuantX for simulated post-training quantization of diffusion pipelines, adding component-aware traversal and configurable linear/convolutional weight and activation granularities for Stable Diffusion 1.5 and SDXL, with exploratory SD3.5 support.
- Built diffusion-trajectory calibration and activation-statistics infrastructure and implemented SmoothQuant-style equivalent scaling; studied PTQ4DM, Q-Diffusion, ViDiT-Q, and SVDQuant to guide the design.
- Diagnosed catastrophic W4 degradation: excluding the convolution path reduced LPIPS from 0.5713 to 0.3380, while subsequent smoothing improved it to 0.3357 under the recorded evaluation protocol.
- Benchmarked TorchAO, Quanto, and QuantX and identified sample-, dataset-, compilation-, and random-state-related evaluation pitfalls; documented the work in a public technical report.

## Academic Service

**Reviewer, NeurIPS 2026 Workshop on Continual World Models** September 2026

*Selected by the workshop Technical Program Committee*

- Reviewed a workshop submission, assessing what the paper demonstrated and where it needed more evidence, and submitted a written review and recommendation to the committee.

## Teaching Experience

**Teaching Assistant, Computer Systems and Assembly Language (CS225)** Fall 2026

*LUMS* *approx. 350 students, two sections*

- Develop quizzes, assignments, and a new systems laboratory; grade examinations and conduct tutorials.

**Teaching Assistant, Foundations of Data Science** Spring 2026

*LUMS* *approx. 200 students*

- Designed introductory Python and data-visualization assignments, automated tests, and Jupyter/Otter-Grader workflows; held office hours, supported course projects, graded work, and invigilated examinations.

**Teaching Assistant, Introduction to Programming (CS100)** Spring 2026

*LUMS* *approx. 180 students*

- Created assignments, held programming labs and a tutorial, and provided student-facing office hours.

## Honors and Awards

- **Dean's Honor List**, LUMS: selected in every completed academic year (2023–24, 2024–25, 2025–26).
- **National Mathematics Talent Contest**: top 50 chosen from 4000+ candidates.
- **Purple Comet! Math Meet**: first-place team in Pakistan.
- **Southeast Asian Mathematical Olympiad (SEAMO)**: Gold Medal; third nationally in Pakistan.
- **Full A-Level Scholarship**: 100% tuition scholarship.

## Technical Skills

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| **Programming:** | C++, Python, C; Bash; Git; Linux; Docker; Jupyter; LaTeX. |
| **ML systems:** | PyTorch; TVM; TensorRT; model quantization; diffusion models; performance evaluation and autotuning. |
| **Architecture:** | Ramulator/Newton PIM; DRAM architecture; heterogeneous GPU/PIM execution; memory-system simulation. |
| **Research methods:** | Empirical evaluation and benchmarking; semi-structured interviews; reflexive thematic analysis. |
| **Languages:** | English, Urdu, Punjabi. |